

Program: Diploma in Renewable Energy	
Course Code : 6291A	Course Title: Distributed Generation
Semester : 6	Credits : 4
Course Category : Program Elective Course	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per Semester : 60

Course Objectives:

- To impart basic ideas of distributed generation and its integration on the grid.
- To understand the fundamental concepts of smart grid, its components and different smart grid communication technologies.
- To familiarize the control and protection issues of microgrids and to understand the concepts of demand side energy management

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic concepts of Electrical & Electronics Engineering	2031	Fundamentals of Electrical & Electronics Engineering	2
Basics concepts of Different renewable sources	2321	Basic Electrical Engineering & Renewable Technology.	2

Course Outcomes :

On completion of the course, the students will be able to:-

CO n.	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the basic concepts of distributed generation and distributed energy resources.	14	Understanding
CO2	Summarize the basic concepts of smart grid and different technologies	16	Understanding

CO3	Explain the various control strategies and different protection issues of microgrids	14	Understanding
CO4	Identify the various concepts related to Demand Side Management	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	3						

3- Strongly mapped 2- Moderately mapped 1- Weakly mapped

Course Outline

On completion of the course, the students will be able to:-

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1: Summarize the basic concepts of distributed generation and distributed energy resources.			
M1.01	Outline the basic ideas for integration of distributed generation to grid.	2	Understanding
M1.02	Describe the basic concept of microgrid and its components	4	Understanding
M1.03	Explain different distributed energy resources	4	Understanding
M1.04	Choose proper storage systems for microgrid applications	4	Understanding

Contents:

Distributed generation – Introduction - Integration of distributed generation to Grid – Active and passive distribution network-Integration issues of distributed generation

Microgrids- Microgrid Concept –Components –AC Microgrid systems-DC Microgrid systems-Comparison of AC and DC Microgrids -Technical , Economic advantage and disadvantages of microgrids- Operational issues of Microgrids

Distributed energy resources: Introduction - Combined heat and power (CHP) systems - Solar photovoltaic (PV) systems – Wind energy conversion systems (WECS) - Small-scale hydroelectric power generation -Other renewable Energy Sources

Energy storage systems (ESS) in microgrid- Different types of Batteries- Advanced lead acid battery, Fuel cell, Flywheel, Supercapacitor, Pumped hydro storage, Superconducting magnetic energy storage, Compressed air energy storage system, Thermal energy storage

CO2	Summarize the basic concepts of smart grid and its different technologies .		
M2.01	Summarize the basic concepts of smart grid and its features.	4	Understanding
M2.02	Summarize the different smart grid interface components.	4	Understanding
M2.03	Explain the different smart grid communication technologies.	4	Understanding
M2.04	Explain the different types of cloud computing in smart grids.	4	Understanding
	Series Test – I	1	

Contents :

Smart Grid- Definitions-Characteristics- Need for smart grid-Functions of smart grid-Evolution of smart Grid-Opportunities of smart grid-Difference between conventional grid and smart grid, Components and architecture- Impacts of smart grid on system reliability-Benefits of smart grid-Features and key challenges of smart grid.

Smart grid interfaces components-storage components-Transmission subsystem components-Monitoring and control technology components- Intelligent and distribution system components-Demand side management component

Smart grid communication technology: Advanced Metering Infrastructure (AMI), Home Area Network (HAN), Neighborhood-Area Networks (NANs), Sensor and Actuator Networks (SANETs) Smart Substations,

Cloud computing in smart grid: Private, public and Hybrid cloud. Cloud architecture of smart grid-Cloud computing in smart grid: Private, Public and hybrid cloud. Types of cloud computing services- Software as a Service (SaaS), Platform as a service (PaaS), Infrastructure as a service (IaaS), Data as a service (DaaS), Cloud architecture for smart grid.

CO3	Explain the various control strategies and protection issues of microgrids		
M3.01	Outline the various control strategies for microgrid.	4	Understanding
M3.02	Summarize the various protection issues for microgrids	4	Understanding
M3.03	Summarize the major protection issues of stand-alone microgrids.	2	Understanding
M3.04	Explain the impact of DG integration on the electricity market	4	Understanding

Contents :

Control of microgrids-Network management requirements of micro grid:Microsource generation control-Domestic process control-Microsource controller:Active and Reactive power control-voltage control- Load sharing through p-f control-Central controller:Energy management module-Protection coordination module

Protection issues for Microgrids: Introduction, Islanding, Different islanding scenarios: Quick separation from faulty feeder-Prevention from false separation-Separation in non-fault conditions-Separation of exporting microgrids-Re-synchronization

Major protection issues of stand- alone Microgrid :Fault protection distribution systems-Protection of microsources

Impact of DG integration on electricity market: Impact on power quality and reliability-Impact on electricity market-Impact on environment-Impact on distribution system-Impact on communication standards and protocols

CO4	Identify the various concepts related to Demand Side Management		
M4.01	Illustrate smart meters and Dynamic pricing	4	Understanding
M4.02	Outline various components in Smart Grid	3	Understanding
M4.03	Summarize basic parameters related to demand side energy management(DSM).	4	Understanding
M4.04	Summarize basic methodologies of load shaping objectives.	3	Understanding
	Series Test – II	1	
Contents : Smart Meters - Introduction to smart meters -Block diagram - Features- Advantages- Disadvantages- Benefits. Dynamic Pricing-Basic concept-Variou s methods:Critical peak pricing,Time of use pricing,Real time pricing,Peak Load Reduction credits. Tariff- Basic concepts -objectives-Availability based tariff,Time of day tariff-Simple problems Smart sensor-Block Diagram-Components-Applications Phasor measurement unit(PMU)- Basic concepts- benefits-Block Diagram of PMU- Wide area measurement system- Basic concepts Demand side and energy management:Role of technology in demand response- Demand Side Management -Load Curves-Load duration curve- Other terms and Factors:Dump power-Firm power-Prime power-Cold reserve-Hot reserve-Spinning reserve Load Shaping Objectives-Methodologies - Peak clipping -Valley filling-Load shifting- Strategic conservation-Strategic load growth-Flexible load shape			

Text / Reference

T/R	Book Title/Author
T1	Dr.Dileep G,A Textbook on 'Distributed Generation and Smart Grid'- Advay Publications,First Edition.
T2	H. Bevrani, B. François, T. Ise, "Microgrid Dynamics and Control", John Wiley & Sons, 1st Edition, 2017
T3	N. D. Hatziargyriou, "Microgrids Architecture and control", IEEE Press Series, John Wiley & Sons Inc, 1st Edition, 2013
R1	S. Chowdhury, S P Chowdhury and P Crossely, "Microgrids and active distribution networks", IET Renewable energy series 6.
R2	Suleiman M. Sharkh, Mohammad A. Abusara, "Power electronic converters for microgrid", IEEE Wiley
R3	Amirnaser Yezdani, and Reza Iravani, Voltage Source Converters in Power Systems: Modeling, Control and Applications, IEEE John Wiley Publications, 2009.
R4	Magdi S. Mahmoud, Microgrid: Advanced Control Methods and Renewable Energy System Integration, Elsevier,2017
R5	Clark W.Gellings, "The Smart Grid: Enabling Energy Efficiency and Demand Response", CRC Press.
R7	James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.

Online Resources

Sl.No	Website Link
1	https://www.nrel.gov/grid/distribution-integration.html
2	https://youtu.be/_W98VjtLjME
3	https://youtu.be/8oPw6uDQRbg
4	https://youtu.be/hav_I2y1YZw
5	https://youtu.be/5uSOp9tvVrA

6	https://youtu.be/bNaWRUVdzTQ
7	https://youtu.be/mf_lKoXVRQo
8	https://youtu.be/WkKzXmwMDBE
9	https://youtu.be/mbQ9ki8FVLI
10	https://youtu.be/qLDOUMBj4as
11	https://youtu.be/uvSr-98u82o
12	https://youtu.be/QKKzmNMhZ-U